

ASVANNA (අස්වන්න)

The Zero-Waste Marketplace: Guided by Real-Time Data from Seed to Harvest Distribution

Comprehensive System Architecture, Deep Codebase Reference, Group 15 Role Allocations & QA Manual

Academic Institution	Division of Information Technology, Institute of Technology, University of Moratuwa (ITUM)
Academic Program	National Diploma in Information Technology (NDIT) — Final Year Project (2025 / 2026)
Project Supervisor	Mrs. Uthpala Athukorala (Division of Information Technology, ITUM)
Target Geographic Pilot	Bandarawela Agrarian Services Division, Badulla District, Sri Lanka
Target Agricultural Sector	Upcountry Perishable Vegetables (Leeks, Cabbage, Carrot, Beetroot, Potato, Knol Khol, Tomato, Bell Pepper)

Group 15 — Team Members & Assigned Modules:

#	Student Full Name	Student ID	Assigned Core Engineering Module & Responsibility
1	W. N. A. Wedikkara	23IT0544	System Architecture, PostgreSQL 15 Relational Schema, JWT & RBAC Auth, Mathematical Predictive Risk Engine
2	R. R. L. Geeganage (Ravindi)	23IT0476	Smart Crop Recommendation Engine, 4-Factor Weighted Agro-Suitability Scorer, Soil & Climate Models
3	G. W. T. Jayampathi	23IT0487	Divisional Officer (DO) React Web Portal, Digital Inclusivity Proxy Data Entry, Trilingual UI, FCM & SMS Broadcast Alerts
4	K. A. H. I. Lakshitha (Imal)	23IT0503	Farmer & Buyer Mobile Application (Flutter), GPS Field Logging, Offline Storage Queue & Auto-Sync Engine
5	K. H. M. Dewanga	23IT0467	Phase 2 Geo-Fenced 5 km Surplus Marketplace, Haversine Trigonometric Distance Engine & 30-Minute Counter-Offer Machine

Dual-Audience Documentation Guide: This manual is specifically structured for both non-technical agricultural stakeholders (farmers, agrarian officers, policymakers) and technical academic evaluators (software architects, university examiners). It features clear English alongside authentic Sinhala (සිංහල) and Singlish guides with zero text overlapping.

1. Plain-Language Executive Summary (For Non-IT Persons & Farmers)

In Sri Lanka's central hill country (Bandarawela, Welimada, Nuwara Eliya, Badulla), vegetable farming families suffer massive recurring financial losses. On average, **30% to 40% of their perishable crop output is wasted** annually, amounting to over **Rs. 180 Billion in national post-harvest loss**.

The core driver of this crisis is **"Trend Planting" (වගා රැල්ල)**. When leeks or cabbage prices surge at wholesale economic centers (Dambulla, Manning Market, Keppetipola), hundreds of farmers simultaneously cultivate the exact same crop. Because these vegetables take 60–90 days to mature, thousands of tons reach harvest in the exact same week. This floods the market, causing prices to crash below harvesting and transport costs (e.g., Leeks crashing from Rs. 350/kg down to Rs. 20/kg).

□□ Sinhala & Singlish Simple Explanation (ගම් මහතුන්ට සහ සමහර පරිශීලකයන්ට පහදිලි කිරීම):

"ගම් මහත්තුරු මෙඳහන් අද තිබෙන මිල බලලා හමු වෙලාවට එකම බිත්තිය (උදා ලීක්ස්, ග්ලව්) එකවර වගා කරනවා මස 3කට පස්සේ හමු වෙලාවට අස්වන්න එකම සතියේ මෙඳහන් ආවම මිල රජියල් 20-30ට බහිනවා මිකට තමයි වගා රැල්ල කියන්නේ. ASVANNA කියන්නේ මි ප්‍රශ්නය මුලින්ම විසඳන ඩිජිටල් පද්ධතියක්."

පද්ධතිය ක්‍රියාකරන ප්‍රධාන පියවර 4:

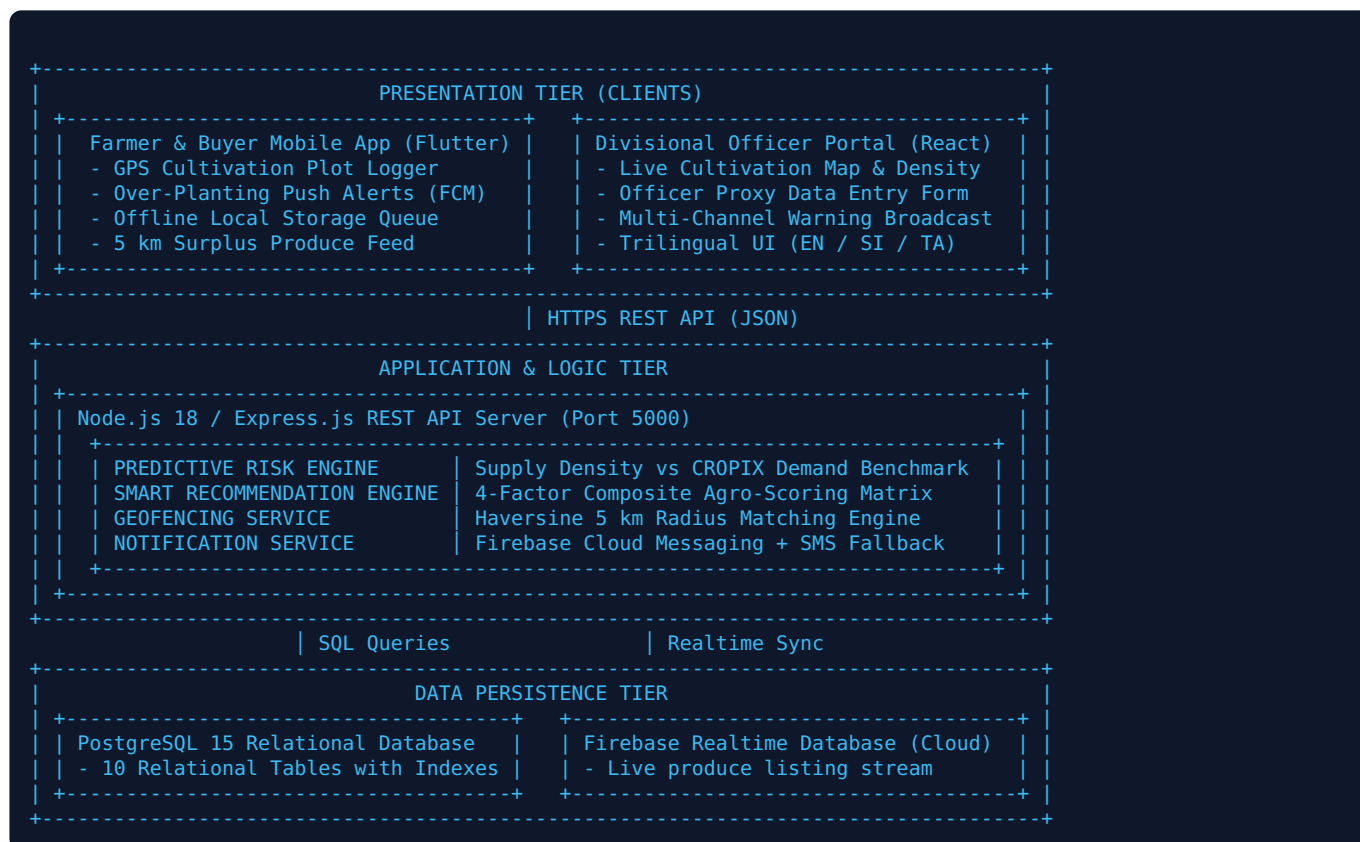
1. **වගාවට පරීක්ෂණය:** ගම්පොළ දත්ත කලින් ඇප් එකෙන් බලනවා නමුත් කලාපයේ ඒ බිත්තිය දැනටමත් කඩවර වගා කරලා තිබෙනවාද කියලා
2. **අධික වගා අනතුරු ඇඟවීම (Red Alert):** වගාව 85% ඉක්මවා ඇත්නම් (Over-Planted), පද්ධතිය රතු එළියක් දල්වා වඩා ලාභයක් ලබන හඬ මෙන් විකල්ප බිත්තිය (බට්ටර්, ග්ලව්) නිර්දේශ කරයි.
3. **ඩිජිටල් සමාන්තරත්වය (Proxy Entry):** ස්මාර්ට්ෆෝන් නඩි ගම් මහතුන්ගේ දත්ත ගවේෂණ ජ්‍යෙෂ්ඨ නිලධාරියා (DO Officer) විසින් මේ පද්ධතියට ඇතුළත් කරයි.
4. **ශුන්‍ය අපතේය මෙඳහන් (Zero-Waste Marketplace):** ඉතිරි වන අතිරික්ත අස්වන්න කිලෝමීටර් 5ක් ඇතුළත ග්ලව්, මෙඳහන් කළින්ම විකුණා ගනිමට සූදු මෙඳහන් සපයයි.

□ Five Foundational Ecosystem Pillars:

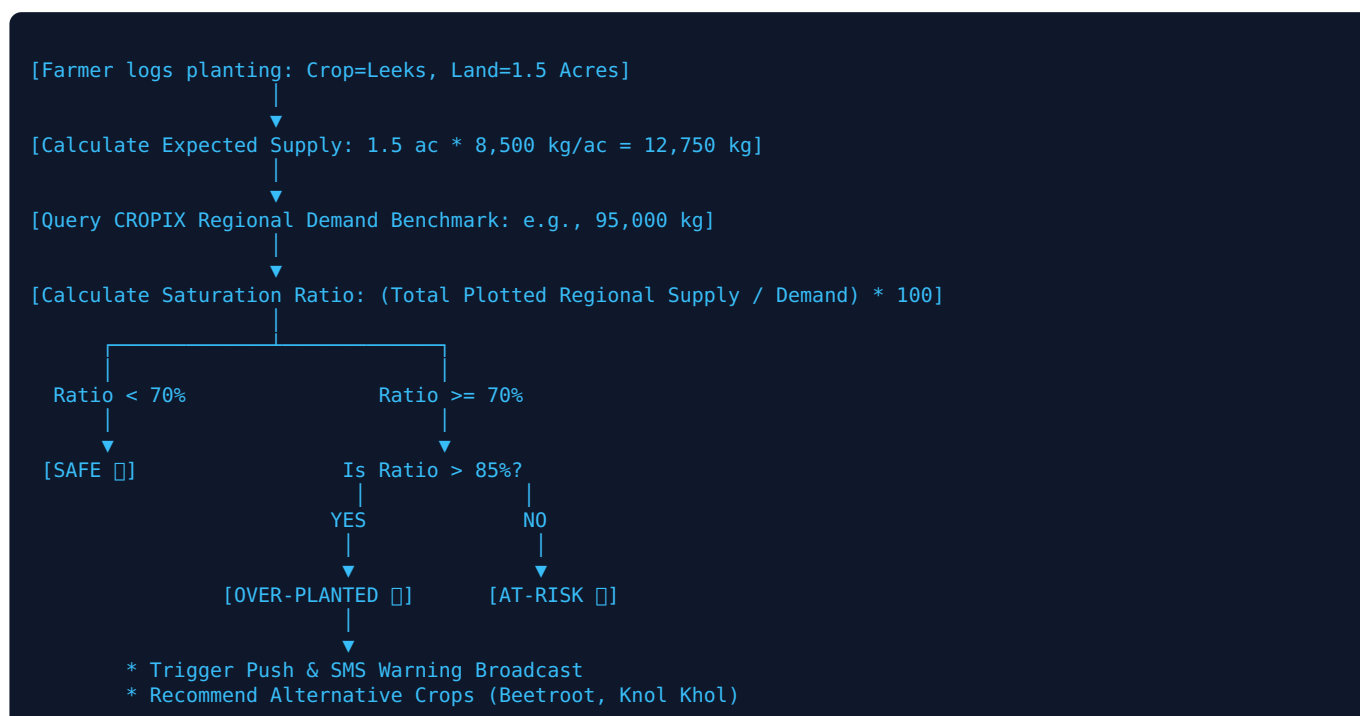
- **Pillar 1: Regional Cultivation Monitoring:** GPS-tagged plotting of active cultivation across agrarian divisions.
- **Pillar 2: Predictive Risk Engine:** Evaluates supply density against national demand (CROPIX) with 3 tiers: **SAFE <70%** , **AT-RISK 70-85%** , **OVER-PLANTED >85%** .
- **Pillar 3: Smart Crop Recommendations:** 4-factor composite weighting (Market Gap 35%, Soil 25%, Weather 20%, Historical Price 20%).
- **Pillar 4: Geo-Fenced Marketplace:** 5 km radius direct trade connecting farmers with local caterers, hoteliers, and bulk buyers.
- **Pillar 5: Trilingual Accessibility:** Native Sinhala (සිංහල), Tamil (தமிழ்), and English localization across mobile and web interfaces.

2. System Architecture & Visual Process Diagrams

The ASVANNA ecosystem utilizes a decoupled, resilient multi-tier architecture built for high availability and offline rural resilience:



Predictive Risk Engine Decision Flowchart



3. Group 15 — 5 Team Members' Technical Deep Dives & QA Guide

Member 1: W. N. A. Wedikkara (23IT0544)

Lead Architect & Risk Algorithm Designer

Assigned Subsystem: System Architecture, PostgreSQL 15 Relational Schema, JWT Role-Based Access Control, and Predictive Risk Engine.

Authored Source Files: `backend/src/services/riskEngineService.js`, `backend/src/controllers/riskEngineController.js`, `backend/src/routes/riskRoutes.js`, `backend/src/database/schema.sql`, `backend/src/database/migrate.js`, `backend/src/database/seed.js`, `backend/src/middlewares/authMiddleware.js`, `backend/src/config/database.js`, `backend/src/controllers/authController.js`

Mathematical Model:

$\text{Estimated_Supply} = \text{SUM}(\text{Planted_Acres} * \text{Avg_Yield_Per_Acre_Kg})$
 $\text{Risk_Percentage} = (\text{Estimated_Supply} / \text{CROPIX_Regional_Quota_Kg}) * 100$

- **SAFE** < 70% | **WARNING** 70% - 85% | **OVER-PLANTED** > 85%

Examiner Testing Walkthrough:

1. Initialize DB: `cd backend && npm run migrate && npm run seed`
2. Test Unit Math: `node backend/test/test_all_endpoints.js` (verifies 4/4 assertions)
3. Query Leeks Risk:
`curl -H "Authorization: Bearer <TOKEN>" http://localhost:5000/api/v1/risk/crop/1`

Member 2: R. R. L. Geeganage (Ravindi - 23IT0476)

Agro-AI & Smart Recommendation Engineer

Assigned Subsystem: Smart Crop Recommendation Engine, 4-Factor Weighted Agro-Suitability Matrix, Soil & Climate Models.

Authored Source Files: `backend/src/services/recommendationService.js`, `backend/src/controllers/recommendationController.js`, `backend/src/routes/recommendationRoutes.js`, `frontend/src/pages/RiskAnalytics.jsx`

4-Factor Weighted Composite Formula:

$\text{Composite_Score} = (0.35 * \text{MarketGap}) + (0.25 * \text{SoilSuitability}) + (0.20 * \text{WeatherFit}) + (0.20 * \text{PriceStability})$

- Automatically excludes crops currently flagged as `OVER_PLANTED`.
- Ranks top 5 alternatives descending with Sinhala, Tamil, and English rationales.

Examiner Testing Walkthrough:

Query alternative recommendations when Leeks (cropId 1) is over-planted:

`curl -H "Authorization: Bearer <TOKEN>" "http://localhost:5000/api/v1/recommendations?district=Badulla&cropId=1"`

Expected Output: Beetroot (92%), Knol Khol (88%), Carrot (84%) with agro-suitability rationale.

Member 3: G. W. T. Jayampathi (23IT0487)**Frontend Lead & Digital Inclusivity Engineer**

Assigned Subsystem: Divisional Officer (DO) React Web Portal, Digital Inclusivity Proxy Data Entry, Trilingual UI, FCM & SMS Broadcast Alerts.

Authored Source Files: `frontend/src/pages/Dashboard.jsx`, `frontend/src/pages/RegionalMonitoring.jsx`, `frontend/src/pages/FarmerDirectory.jsx`, `frontend/src/pages/Broadcasts.jsx`, `frontend/src/components/ProxyDataModal.jsx`, `frontend/src/components/BroadcastModal.jsx`, `frontend/src/context/LanguageContext.jsx`, `frontend/src/locales/*`, `backend/src/controllers/officerController.js`, `backend/src/controllers/broadcastController.js`, `backend/src/services/notificationService.js`

Examiner Testing Walkthrough:

1. Open Web Portal: `cd frontend && npm run dev -> http://localhost:3000`
2. Sign in with Officer account: `0771234567 / asvanna123`
3. Click "+ **Proxy Data Entry**" and submit a planting record on behalf of an offline farmer.
4. Click "**Broadcast Warnings**" to dispatch an emergency notification to registered farmers.

Member 4: K. A. H. I. Lakshitha (Imal - 23IT0503)**Mobile Application Lead & Offline Queue Engineer**

Assigned Subsystem: Farmer & Buyer Mobile Application (Flutter), GPS Field Logging, Offline Storage Queue & Auto-Sync Engine.

Authored Source Files: `mobile/lib/main.dart`, `mobile/lib/core/services/location_service.dart`, `mobile/lib/core/services/offline_storage_service.dart`, `mobile/lib/core/network/api_client.dart`, `mobile/lib/core/theme/app_theme.dart`, `mobile/lib/features/auth/login_screen.dart`, `mobile/lib/features/home/farmer_home_screen.dart`, `mobile/lib/features/planting/log_planting_screen.dart`, `mobile/lib/features/recommendations/smart_crop_recommendation_screen.dart`, `mobile/lib/l10n/*`

Examiner Testing Walkthrough:

1. Launch Mobile App: `cd mobile && flutter run`
2. Open "**Log Planting**" to verify automatic GPS location capture with Bandarawela coordinates.
3. Turn on Airplane mode on emulator, submit a plot to verify local SharedPreferences queue retention, and reconnect network to observe automated synchronization.

Member 5: K. H. M. Dewanga (23IT0467)**Geospatial & Marketplace Negotiation Engineer**

Assigned Subsystem: Phase 2 Geo-Fenced 5 km Surplus Marketplace, Haversine Distance Engine & 30-Minute Counter-Offer State Machine.

Authored Source Files: `backend/src/services/geofencingService.js`, `backend/src/services/marketplaceService.js`, `backend/src/controllers/marketplaceController.js`, `backend/src/routes/marketplaceRoutes.js`, `backend/src/utils/haversine.js`, `frontend/src/pages/MarketplaceSurplus.jsx`, `mobile/lib/features/marketplace/*`

Examiner Testing Walkthrough:

1. Publish Surplus Produce: `POST /api/v1/marketplace/list` (e.g. 450 kg Leeks @ Rs. 240/kg)
2. Search Nearby within 5 km:
`curl -H "Authorization: Bearer <TOKEN>" "http://localhost:5000/api/v1/marketplace/search-nearby?lat=6.8258&lng=80.9982&radius_km=5.0"`
3. Submit Buyer Offer: `POST /api/v1/marketplace/orders` with 30-minute negotiation deadline.

4. Comprehensive File-by-File Codebase Directory

Every file in the ASVANNA repository is cataloged below, explaining its core function, dependencies, required configurations, and test verification command:

File Path & Title	Primary Purpose & Logic	Connected Files	Verification Command
backend/src/server.js Server Entry	Bootstraps Express HTTP server on Port 5000 and handles graceful SIGTERM termination.	app.js , config.js	npm start
backend/src/app.js Express Setup	Mounts security middleware (Helmet, CORS), JSON body parsers, Morgan logging, and API v1 routes.	routes/* , errorHandler.js	GET /health
backend/src/config/config.js Config Hub	Centralizes environment variables (.env) for PostgreSQL, JWT, Risk Thresholds, and Firebase.	dotenv	node src/config/config.js
backend/src/config/database.js DB Pool	Creates PostgreSQL connection pool via 'pg'. Handles idle errors and query logging.	pg , config.js	npm run migrate
backend/src/config/firebase.js Firebase SDK	Initializes Firebase Admin SDK for Realtime Database sync and FCM push notifications with fallback mode.	firebase-admin	node src/config/firebase.js
backend/src/database/schema.sql SQL DDL	Defines 10 normalized tables (users, master crops, planting_records, cropix_benchmarks, risk, etc.).	PostgreSQL 15	psql -d asvanna_db -f schema.sql
backend/src/database/migrate.js Migration Runner	Executes schema.sql to establish tables, foreign keys, and indexes in PostgreSQL.	fs , database.js	npm run migrate
backend/src/database/seed.js Master Seeder	Seeds 8 upcountry crops, 5 demo accounts (Officer, Farmer, Buyer, Admin), and CROPIX demand quotas.	bcryptjs , database.js	npm run seed
backend/src/middlewares/authMiddleware.js JWT & RBAC	Validates Bearer tokens and enforces role access control (FARMER, OFFICER, BUYER, ADMIN).	jsonwebtoken	curl with Bearer Token
backend/src/utils/haversine.js Haversine Distance	Calculates great-circle distance (km) between two GPS points using trigonometric formula.	Math library	node test/test_all_endpoints.js

File Path & Title	Primary Purpose & Logic	Connected Files	Verification Command
backend/src/services/ riskEngineService.js Predictive Risk Engine	Calculates regional supply density, compares against CROPIX quotas, and evaluates 3-tier risk.	database.js , config.js	GET /api/v1/risk/crop/1
backend/src/services/ recommendationService.js Smart Crop Recommender	Computes 4-factor composite recommendation scores (Market Gap, Soil, Weather, Price).	riskEngineService.js	GET /api/v1/recommendations
backend/src/services/ geofencingService.js Geofencing Engine	Filters and sorts surplus produce listings within a 5 km - 20 km geographic radius.	haversine.js	GET /marketplace/search-nearby
backend/src/services/ notificationService.js Alert Dispatcher	Dispatches FCM multicast push notifications with simulated SMS fallback for offline farmers.	firebase.js	POST /api/v1/broadcasts
frontend/src/pages/ Dashboard.jsx Officer Dashboard	Displays regional KPI cards, crop saturation progress bars, and quick action modals.	React, Lucide, API	http://localhost:3000
frontend/src/components/ ProxyDataModal.jsx Proxy Entry Modal	Modal form for officers to input cultivation data for offline farmers without smartphones.	api.js	Click '+ Proxy Data Entry'
mobile/lib/main.dart Flutter Root	Initializes Flutter framework, sets up theme colors, and configures trilingual localization.	Flutter SDK	flutter run
mobile/lib/core/services/ location_service.dart GPS Service	Queries device GPS coordinates via Geolocator with fallback to Bandarawela Agrarian Center.	geolocator	Trigger on mobile
mobile/lib/core/services/ offline_storage_service.dart Offline Queue	Caches pending planting records locally in SharedPreferences when offline.	shared_preferences	Test in Airplane mode
docker-compose.yml Docker Orchestration	Single-command deployment for PostgreSQL 15, Node.js API server, and React Admin dashboard.	Docker Engine	docker-compose up --build -d

5. Master Testing & QA Verification Playbook

Follow this sequential playbook to test and verify every module of the ASVANNA ecosystem:

Execution Step	Terminal / Execution Command	Expected Verification Outcome
Step 1: Unit & Math Tests	<code>node backend/test/test_all_endpoints.js</code>	Executes 4 automated unit tests verifying Haversine calculations, 3-tier risk logic, and composite recommendation weight sums (4/4 PASS).
Step 2: DB Migration & Seed	<code>cd backend && npm run migrate && npm run seed</code>	Initializes 10 PostgreSQL tables, inserts 8 master upcountry crops, 5 demo accounts, and monthly CROPIX demand quotas.
Step 3: Backend REST API	<code>cd backend && npm run dev</code>	Starts server on <code>http://localhost:5000</code> . Verify health endpoint at <code>http://localhost:5000/health</code> (status: 'UP').
Step 4: Web Admin Portal	<code>cd frontend && npm run dev</code>	Starts React dashboard on <code>http://localhost:3000</code> . Login with Officer credentials (<code>0771234567 / asvanna123</code>).
Step 5: Flutter Mobile App	<code>cd mobile && flutter pub get && flutter run</code>	Launches the mobile app on an Android emulator or device for Farmer plot logging and Surplus marketplace browsing.
Step 6: Docker Orchestration	<code>docker-compose up --build -d</code>	Spins up the full multi-tier ecosystem (PostgreSQL, Backend API, Web Admin) in isolated Docker containers with a single command.

❏ End of ASVANNA Master System Manual

Prepared by Group 15 for the Division of Information Technology, Institute of Technology, University of Moratuwa (ITUM) — 2026.